DEEP LEARNING THEORY–BASED PRACTICAL WORK

Phase 2 · Solution Design

Cross-Subject Motor Imagery EEG Classification using Deep Learning

A hybrid convolutional–Transformer approach evaluated on unseen subjects

Team members	Ananya Gourishetty · Haarika D · Siddhanth Samal · Sathvik P
Guide / faculty	Dr. Mohan Allam
Course	CSE4006 – Deep Learning · Group 5
Institution	School of Computer Science and Engineering, VIT-AP University
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PROBLEM STATEMENT

Decoding imagined movement from scalp EEG

When a person imagines moving a limb, the sensorimotor cortex shows a measurable drop in mu-band power on the opposite side of the head. The task is to recover which movement was imagined from that signal.

Low signal-to-noise ratio

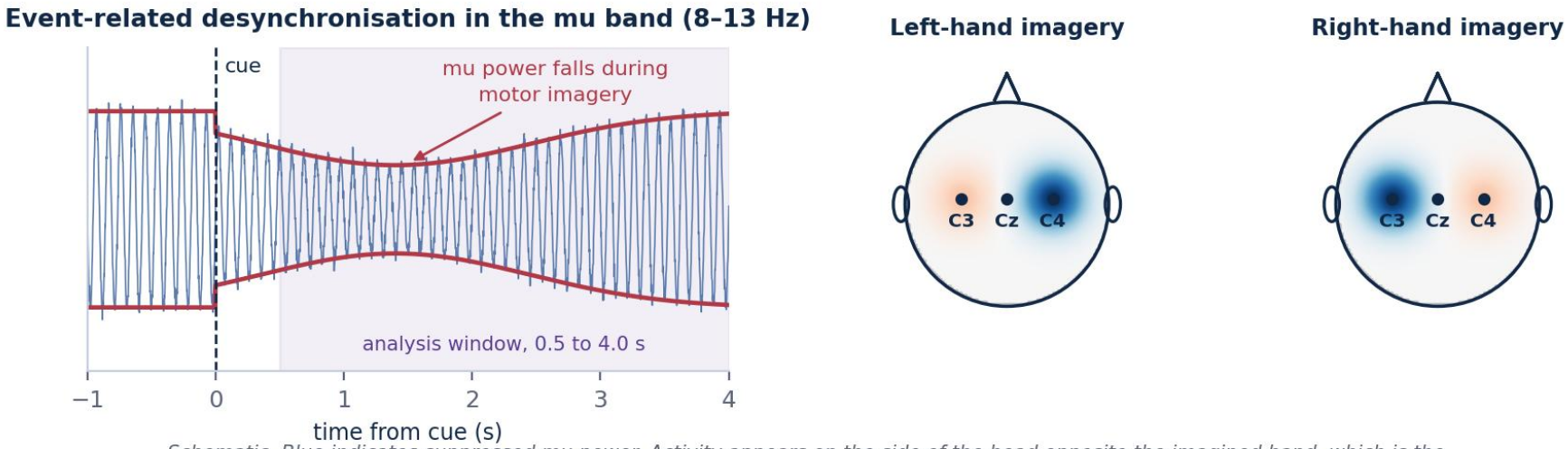
Cortical activity is attenuated by the skull and scalp, and is mixed with eye, muscle and mains interference

Small, high-dimensional data

A four-second trial holds over twenty thousand samples, but each subject provides only a few hundred labelled trials

Variability between people

Skull shape, electrode placement and mental strategy differ, so feature distributions shift from subject to subject



Schematic. Blue indicates suppressed mu power. Activity appears on the side of the head opposite the imagined hand, which is the pattern a correctly trained decoder is expected to rely on.

LITERATURE SURVEY SUMMARY

From compact CNNs to convolutional–Transformer hybrids

2018

EEGNet

Compact CNN

Depthwise and separable convolution

2,548 parameters

2023

ATCNet

CNN + attention + TCN

Sliding-window attention over temporal features

113,732 parameters

2023

EEG Conformer

CNN + Transformer

Convolution then global self-attention

78.66% within-subject

2024

CTNet

CNN + Transformer

EEGNet front end, 6-layer Transformer encoder

82.52% within-subject

Increasing capacity to model long-range temporal structure

Increasing parameter count and data requirement

What improved

Each step added a better way to relate information across the whole trial, rather than only within a short window.

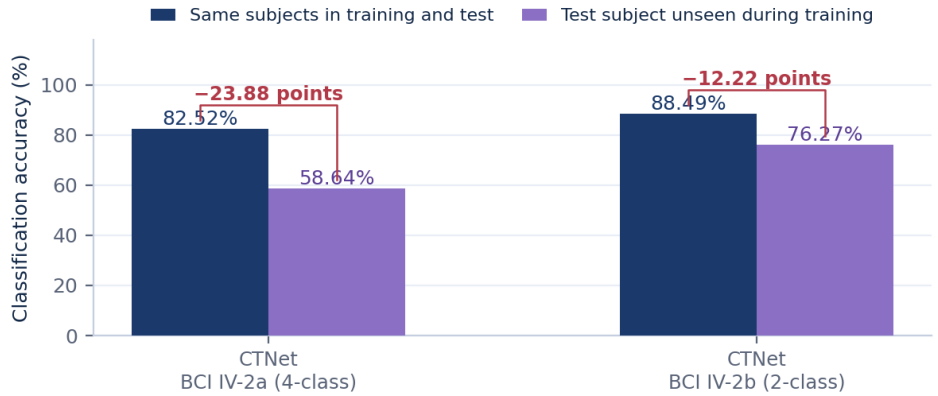
What it cost

Parameter counts rose from about two and a half thousand to over one hundred thousand, and the data requirement rose with them.

RESEARCH GAP

Four limitations common to the surveyed work

1. Cross-subject generalisation remains difficult; accuracy falls when the test subject is unseen.
2. Zero-calibration and low-calibration operation is still an open problem.
3. CNNs are lightweight but limited in modelling long-range temporal structure.
4. Transformers capture global relationships but cost more and need more data.



Values reported by Zhao et al., Scientific Reports 14:20237 (2024).

What is needed

A model that balances classification accuracy, model complexity and generalisation to new people, rather than optimising any one of them alone.

The clearest illustration

CTNet reaches 82.52% within subject, yet loses 23.88 points on the four-class task when the test subject is unseen.

PROJECT OBJECTIVES

What the project sets out to do

1. Implement a reproducible preprocessing and evaluation pipeline for BCI Competition IV Dataset 2a.
2. Implement EEGNet, ATCNet, CTNet and EEG Conformer as baselines under one identical protocol.
3. Design a hybrid model combining the convolutional front end of EEGNet with the attention mechanisms of the other three.
4. Train and evaluate every model using leave-one-subject-out cross-validation.
5. Measure accuracy, Cohen's kappa, precision, recall and F1-score, per subject and averaged.
6. Compare parameter count and computational cost alongside accuracy.

The final objective is to report whether the proposed combination improves cross-subject performance while remaining lightweight — and to report it either way.

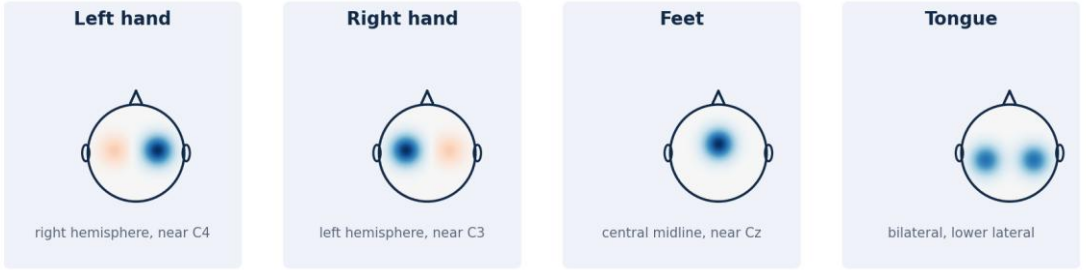
DATASET

BCI Competition IV Dataset 2a

Property	Value
Subjects	9
EEG channels	22
Classes	4 — left hand, right hand, feet, tongue
Sampling rate	250 Hz
Sessions per subject	2, on different days
Trials per subject	576
Total labelled trials	5,184
Analysis window	0.5 to 4.0 s after cue (875 samples)

Chosen because all four surveyed papers evaluate on it, so our baseline implementations can be checked against published values.

The four motor imagery classes

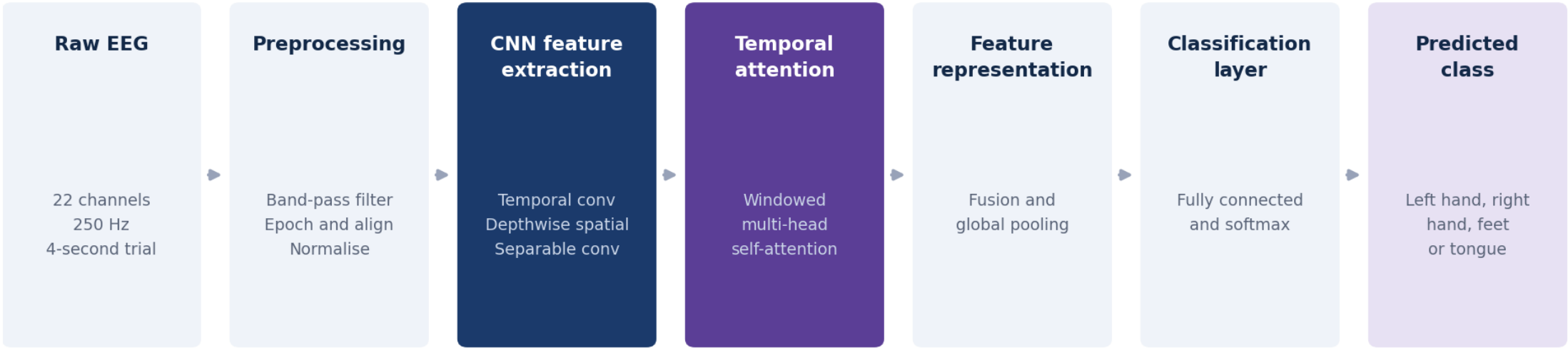


Each class activates a different region of the sensorimotor cortex. Hand imagery appears on the opposite side of the head, foot imagery near the midline.

With only nine subjects, leave-one-subject-out gives nine folds — enough for a mean, a standard deviation and a paired test, but not enough to treat any single fold as reliable on its own.

PROPOSED SYSTEM ARCHITECTURE

From raw EEG to a predicted movement



The two shaded blocks are the learned stages that together form the proposed hybrid model.

CNN feature extraction

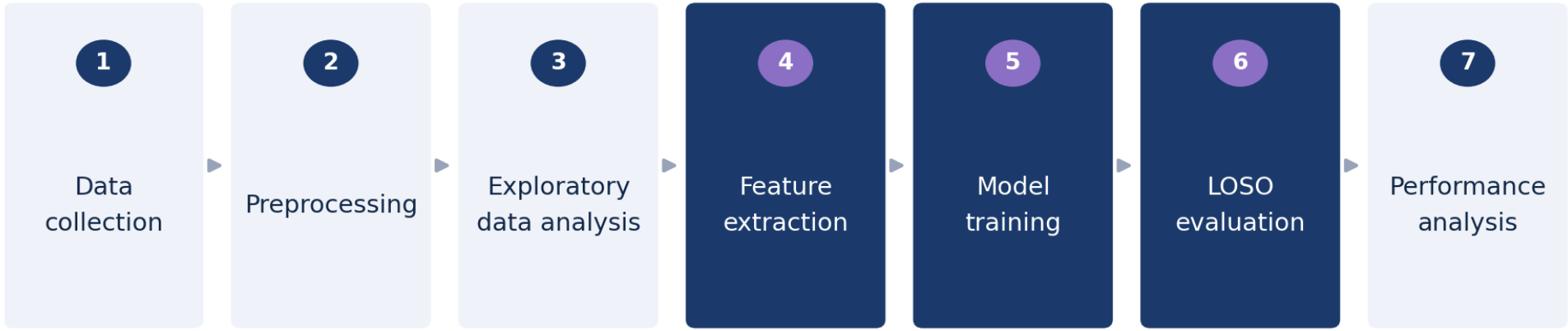
Depthwise and separable convolution, adapted from EEGNet. Learns spatial filters at very low parameter cost.

Temporal attention

Windowed self-attention, adapted from ATCNet, CTNet and EEG Conformer. Relates distant parts of the trial.

PROPOSED METHODOLOGY

Step-by-step pipeline



Stage	What happens	Why it matters
Preprocessing	Band-pass 4–38 Hz, epoch 0.5–4.0 s, alignment, normalisation	Identical for every model, so differences come from the models
Feature extraction	Convolutional block followed by windowed attention	The learned part of the system
Model training	Subject-independent, validation on a held-out subject	Avoids selecting hyperparameters that favour within-subject fit
LOSO evaluation	Nine folds, each with a different unseen test subject	Measures what a new user would experience
Performance analysis	Metrics per subject, plus a paired statistical test	Averages alone hide failures on individual subjects

BASELINE COMPARISON

Four published baselines and the proposed model

Model	Architecture type	Main strength	Main limitation	Role in our experiments
EEGNet (2018)	Compact CNN	Very small; trains reliably on limited data	Limited long-range temporal modelling	Lower bound on model size
ATCNet (2023)	CNN + attention + TCN	Attention applied locally, suits short trials	Considerably larger than EEGNet	Source of the windowed attention idea
EEG Conformer (2023)	CNN + Transformer	Global self-attention with interpretability	Costly; needs augmentation to train well	Reference for convolution-then-attention
CTNet (2024)	CNN + Transformer	Strong within-subject accuracy (82.52%)	Large drop under subject-independent testing	States the research gap quantitatively
Proposed HCT-Net	CNN + windowed attention	Combines the components above at 20,996 parameters	Not yet evaluated	The model under investigation

Published accuracies follow on the next slide, split by protocol: cross-subject results measured under one shared protocol, and within-subject figures that come from different protocols and cannot be ranked against each other.

Published results, split by protocol

Cross-subject: one protocol for all (Zhao et al., 2024)

Model	Unseen person	Kappa
ShallowConvNet+	56.75 ± 13.77	0.4234
DeepConvNet+	60.15 ± 12.71	0.4686
EEGNet+	56.85 ± 15.82	0.4246
EEG Conformer+	53.41 ± 17.08	0.3789
CTNet	58.64 ± 14.61	0.4486

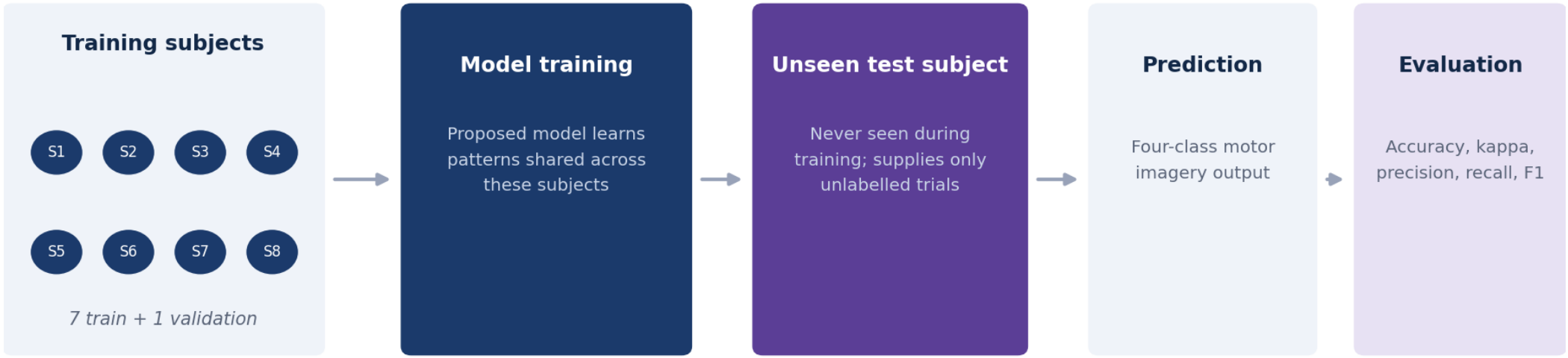
Within-subject: different protocols, not comparable

Model	Same person	Source
CTNet	82.52%	Zhao et al. (2024)
EEG Conformer	78.66%	Authors' repository, hold-out
ATCNet	85.38%	Paper, subject-dependent
ATCNet	81.10%	ATCNet repository script
EEGNet	68.67%	ATCNet repository script
EEGNet+	77.39%	Zhao et al. (2024), re-implemented

Cross-subject: each subject held out in turn, trained on the other eight for 600 epochs on the 2–6 s window, no alignment and no fine-tuning on the held-out subject. + marks baselines re-implemented by the CTNet authors under the same conditions. This is the reference for our own cross-subject results.

EVALUATION STRATEGY

Leave-one-subject-out evaluation



This rotates nine times so that every subject serves as the unseen test subject exactly once.

Accuracy
Mean and standard deviation over nine folds

Cohen's kappa
Corrected for chance agreement

Precision, recall, F1
Macro-averaged over the four classes

Parameters and time
Model complexity and computational cost

A Wilcoxon signed-rank test, paired across the nine subjects, is used to decide whether a difference between two models is meaningful. Results are reported per subject as well as averaged.

EXPECTED OUTCOMES AND NOVELTY

What we expect, and what is actually new

Expected outcomes

- Better generalisation to unseen subjects than the heavier baselines
- Competitive motor imagery classification accuracy
- A lightweight architecture — 20,996 parameters, about one fifth the size of ATCNet
- Reduced dependence on subject-specific calibration

What the project contributes

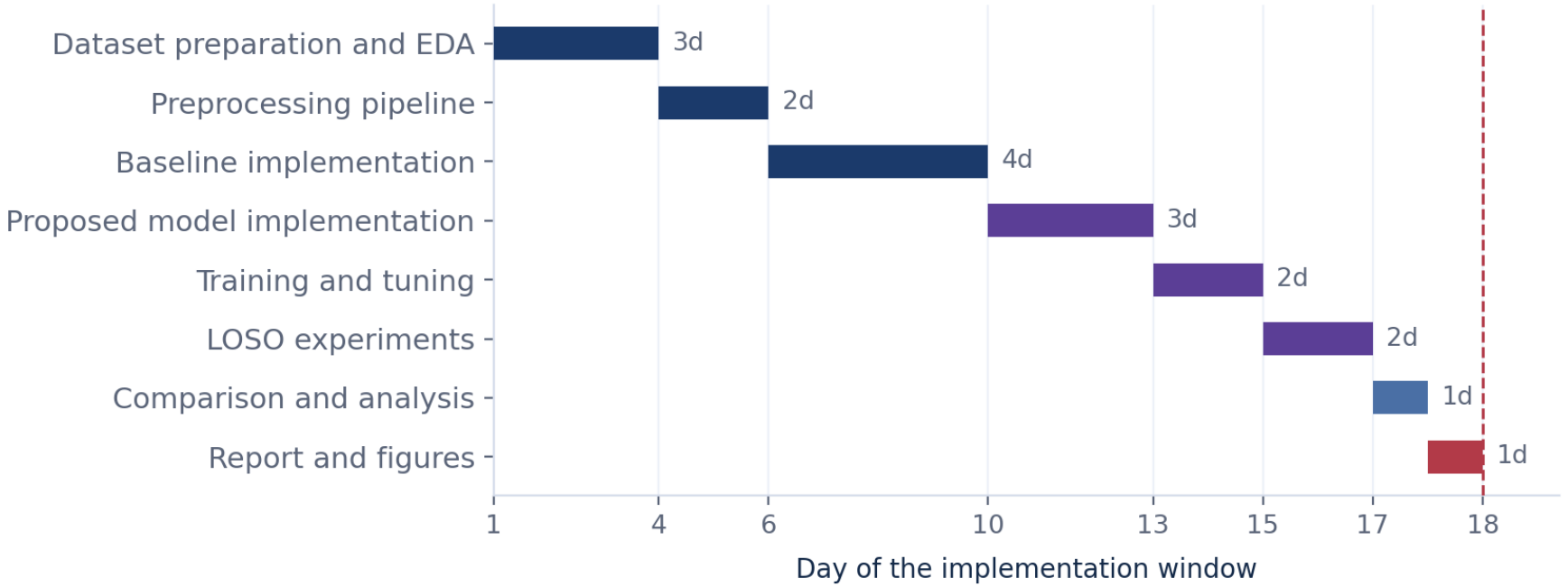
- A comparison of all four architectures under one identical LOSO protocol
- A specific lightweight combination, with attention depth reduced to suit the dataset size
- A component-wise study showing what each part contributes to cross-subject accuracy
- Accuracy measured against parameter count under a single protocol

These are expected outcomes, not results.

No experiments have been run at this stage. The individual components are all taken from published work; what is proposed is the particular combination, the subject-independent evaluation, and the study of what each component contributes.

PHASE 3, 16 SEPTEMBER TO 3 OCTOBER

Implementation plan and conclusion



Durations shown in days; the sequence matters more than the calendar

In summary

The literature survey identified a specific, measurable gap: cross-subject accuracy falls sharply, and closing it currently costs model size. Phase 2 turns that into an implementable plan — one dataset, five models, one evaluation protocol, and a fixed schedule. The plan is deliberately front-loaded, so that the pipeline and baselines are verified before the proposed model is written.